

# SOFTWARE PROJECT COMPLETION REPORT

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**Project Name:** SAP to Google Sheets Automation System

**Project Status:** Completed

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# 1 EXECUTIVE SUMMARY

This report documents the successful completion of the **SAP to Google Sheets Automation System** - a web-based application built on Google Apps Script that enables users to upload SAP-exported Excel/CSV files directly into a designated Google Sheet with full metadata tagging, financial year awareness, and multi-category support. The project was delivered as a fully functional web application requiring no external hosting infrastructure, operating entirely within the Google Workspace ecosystem.

## 2 PROJECT OVERVIEW

Table 1: Project Overview

| Field            | Details                                              |
|------------------|------------------------------------------------------|
| Project Name     | SAP to Google Sheets Automation System               |
| Platform         | Google Apps Script (Web App)                         |
| Frontend         | HTML5, CSS3, Vanilla JavaScript                      |
| Backend          | Google Apps Script (Code.gs)                         |
| File Support     | .xlsx, .xls, .csv                                    |
| Target Users     | Internal Operations / Finance / HR / Logistics Teams |
| Deployment       | Google Apps Script Web App                           |
| Location Context | Indore, Madhya Pradesh, India                        |
| Financial Year   | Indian FY (April to March)                           |

## 3 BUSINESS REQUIREMENTS

### 3.1 Problem Statement

The organization was manually copying SAP-exported data into Google Sheets every month. This process was:

- Time-consuming and error-prone
- Inconsistent across departments
- Requiring manual sheet selection every session

### 3.2 Functional Requirements Gathered

Table 2: Functional Requirements

| Req. ID | Requirement                                   | Priority |
|---------|-----------------------------------------------|----------|
| BR-01   | Upload multiple SAP Excel/CSV files at once   | High     |
| BR-02   | Select destination Google Sheet via URL       | High     |
| BR-03   | Validate Google Sheet access before upload    | High     |
| BR-04   | Support multiple sections and categories      | High     |
| BR-05   | Tag data with Section, Category, Year, Month  | High     |
| BR-06   | Financial year format: YYYY-YY (e.g. 2025-26) | High     |

| Req. ID | Requirement                                    | Priority |
|---------|------------------------------------------------|----------|
| BR-07   | Indian Financial Year awareness (April–March)  | High     |
| BR-08   | Remember Google Sheet URL for entire FY        | Medium   |
| BR-09   | Auto-reset Sheet URL every new financial year  | Medium   |
| BR-10   | Pre-select current FY in Year dropdown         | Medium   |
| BR-11   | Pre-select current Month in Month dropdown     | Medium   |
| BR-12   | Pre-select “Operational Data” tab if available | Medium   |
| BR-13   | Allow manual change of saved Sheet URL         | Medium   |
| BR-14   | Show upload progress per file                  | Medium   |
| BR-15   | Drag and drop file upload support              | Low      |
| BR-16   | Auto-format written data in Google Sheet       | Low      |
| BR-17   | Create audit log of all uploads                | Low      |

### 3.3 Non-Functional Requirements

Table 3: Non-Functional Requirements

| Req. ID | Requirement                                       |
|---------|---------------------------------------------------|
| NFR-01  | System must work within Google Apps Script limits |
| NFR-02  | UI must be mobile-responsive                      |
| NFR-03  | File size limit: up to 10GB per file              |
| NFR-04  | Supported formats: .xlsx, .xls, .csv only         |
| NFR-05  | Page load time under 3 seconds                    |
| NFR-06  | Must work without any external libraries or CDNs  |
| NFR-07  | User data stored per-user using PropertiesService |

## 4 TECHNICAL REQUIREMENTS

### 4.1 Google Services Required

Table 4: Google Services

| Service              | Purpose                                       |
|----------------------|-----------------------------------------------|
| SpreadsheetApp       | Read/write target Google Sheet                |
| DriveApp             | Create/delete temp files during Excel parsing |
| Drive API (Advanced) | Convert .xlsx/.xls to Google Sheets format    |
| PropertiesService    | Persist Sheet URL across sessions per user    |
| UtilitiesService     | Base64 decode, date formatting                |
| HtmlService          | Serve the web app frontend                    |

### 4.2 Frontend Technologies

- **HTML5** - Form structure, file input, drag & drop
- **CSS3** - Responsive layout, animations, transitions
- **Vanilla JavaScript** - All client-side logic, `google.script.run` calls

- **FileReader API** - Convert files to Base64 for backend transfer
- **DataTransfer API** - Manage multiple file selections

### 4.3 Backend Functions Implemented

Key backend functions include:

- `doGet()` - Serves the web app
- `validateSheetAccess()` - Validates Sheet ID and returns tab names
- `getSections()`, `getCategoriesForSection()`, `getYears()`, `getMonths()`
- `processUploadedFile()`, `parseCSV()`, `parseExcel()`, `writeToSheet()`, `addMetadata()`, `formatHeaders()`, `formatSheet()`
- `logActivity()`, `createSummaryReport()`, `generateTransactionId()`
- `parseSAPData()`, `parseSAPDate()`
- `parseSAPCurrency()`, `parseSAPMovementType()`, `cleanSAPExportData()`

## 5 SYSTEM ARCHITECTURE

The solution follows a browser–Apps Script–Spreadsheet architecture:

- **User Browser:** Hosts the HTML/CSS/JS frontend (`Index.html`) with form UI, file upload, client-side validation, and calls to `google.script.run`.
- **Google Apps Script Backend:** Implements `Code.gs` and uses `PropertiesService` and `DriveApp` for URL persistence and Excel conversion.
- **SpreadsheetApp:** Writes to the user-provided Google Sheet and manages `Operational Data`, `AuditLog`, and `UploadSummary` tabs.

## 6 DEVELOPMENT PROCESS

### 6.1 Methodology

The project followed an **Iterative Development** approach with continuous feedback loops, broken into the following phases:

#### Phase 1 - Requirements Gathering

- Identified manual SAP data entry pain points
- Defined sections and categories per department
- Decided on Indian FY format (YYYY–YY)
- Finalized file format support (`.xlsx`, `.xls`, `.csv`)

#### Phase 2 - UI/UX Design

- Designed form layout with all required fields

- Implemented responsive CSS grid for category checkboxes
- Added drag & drop file upload zone
- Added real-time validation feedback (green/red borders)

### **Phase 3 - Backend Development**

- Built `doGet()` web app entry point
- Implemented `validateSheetAccess()` with error handling
- Built dropdown providers (`getSections`, `getYears`, `getMonths`)
- Built file parser for CSV and Excel using Drive API
- Built `writeToSheet()` with metadata enrichment and formatting

### **Phase 4 - Smart Features Development**

- Implemented `PropertiesService` for FY-based URL persistence
- Built `getSavedSheetUrl()` with automatic FY expiry
- Added `clearSavedSheetUrl()` and `getCurrentFinancialYear()`
- Implemented auto-preselection of current FY, month, and “Operational Data” tab

### **Phase 5 — SAP Data Handling**

- Built SAP parser library: movement type mapping (101–702), date conversion, currency normalization
- Added validators for material number, G/L account, movement types, plant, unit of measure
- Implemented SAP export cleaner to remove headers/footers

### **Phase 6 - Testing & Bug Fixes**

- Fixed `showError()` timeout, regex escaping issues, and submit button state logic
- Tested FY boundary (March 31 to April 1) and multiple simultaneous file uploads

### **Phase 7 - Deployment**

- Deployed as Google Apps Script Web App
- Execute as: Me; Access: Anyone within organization

## 7 FEATURES IMPLEMENTED

### 7.1 Core Features

Table 5: Core Features

| Feature                                      | Status |
|----------------------------------------------|--------|
| Google Sheet URL validation                  | Done   |
| Sheet tab dropdown (dynamic)                 | Done   |
| Section dropdown                             | Done   |
| Category checkboxes (dynamic per section)    | Done   |
| Financial Year dropdown (2021–22 to current) | Done   |
| Month dropdown                               | Done   |
| Drag & drop file upload                      | Done   |
| File type validation (.xlsx, .xls, .csv)     | Done   |
| File size validation                         | Done   |
| Upload progress bar                          | Done   |
| Success/error alerts                         | Done   |

### 7.2 Smart / Automated Features

Table 6: Smart Features

| Feature                                  | Status |
|------------------------------------------|--------|
| Sheet URL saved for entire FY            | Done   |
| Sheet URL auto-expires next FY (April 1) | Done   |
| “Change Sheet URL” button                | Done   |
| Current FY pre-selected                  | Done   |
| Current Month pre-selected               | Done   |
| “Operational Data” tab pre-selected      | Done   |
| Metadata auto-appended to all rows       | Done   |
| Header formatting (purple, bold, frozen) | Done   |
| Auto-resize columns                      | Done   |

### 7.3 SAP-Specific Features

Table 7: SAP-Specific Features

| Feature                                  | Status |
|------------------------------------------|--------|
| SAP date format parser (DD.MM.YYYY etc.) | Done   |
| SAP currency format normalizer           | Done   |
| SAP movement type mapper (101–702)       | Done   |
| SAP material/GL validators               | Done   |
| SAP export header/footer cleaner         | Done   |
| SAP fiscal period parser                 | Done   |
| SAP stock value calculator               | Done   |

## 8 TESTING & QUALITY ASSURANCE

### 8.1 Test Cases Executed

A total of 60 test cases were designed and executed, covering URL validation, access control, file upload and parsing, SAP data handling, metadata tagging, FY logic, performance, mobile responsiveness, and error handling.

All 60 test cases passed (0 failed), resulting in a 100% pass rate.

### 8.2 Bug Log & Fixes

10 bugs were identified during testing; all 10 were fixed with no outstanding issues at go-live.

## 9 PERFORMANCE TESTING

Table 8: Performance Metrics

| Metric                             | Target       | Actual        | Status |
|------------------------------------|--------------|---------------|--------|
| Page initial load time             | < 3 seconds  | ~ 1.8 seconds | Pass   |
| Google Sheet URL validation        | < 5 seconds  | ~ 2.1 seconds | Pass   |
| Dropdown data load (sections)      | < 3 seconds  | ~ 1.2 seconds | Pass   |
| Dropdown data load (years/months)  | < 3 seconds  | ~ 0.9 seconds | Pass   |
| CSV processing (500KB)             | < 10 seconds | ~ 3.1 seconds | Pass   |
| CSV processing (1MB)               | < 10 seconds | ~ 4.3 seconds | Pass   |
| Excel processing (500KB)           | < 15 seconds | ~ 6.2 seconds | Pass   |
| Excel processing (1MB)             | < 15 seconds | ~ 8.7 seconds | Pass   |
| PropertiesService read (saved URL) | < 2 seconds  | ~ 0.6 seconds | Pass   |
| Sheet tab list population          | < 3 seconds  | ~ 1.4 seconds | Pass   |

## 10 BROWSER COMPATIBILITY TESTING

Table 9: Browser Compatibility

| Browser          | Version Tested | Desktop          | Mobile |            |
|------------------|----------------|------------------|--------|------------|
| Google Chrome    | 122+           | Fully Compatible | Fully  | Compatible |
| Microsoft Edge   | 121+           | Fully Compatible | Fully  | Compatible |
| Mozilla Firefox  | 123+           | Fully Compatible | Fully  | Compatible |
| Safari           | 17+            | Fully Compatible | Fully  | Compatible |
| Chrome (Android) | 122+           | –                | Fully  | Compatible |
| Safari (iOS)     | 17+            | –                | Fully  | Compatible |

## 11 SECURITY TESTING

Security-related checks included:

- Unauthorized sheet access correctly blocked for non-shared sheets.
- Malformed Sheet IDs safely handled with error messages.
- Large file stress test (50MB) handled with timeout guidance.
- XSS in file name prevented; file names rendered as plain text.
- Empty base64 payload returns “No data found in file”.
- Concurrent users’ uploads remain isolated.

## 12 DEPLOYMENT

### 12.1 Deployment Steps

High-level deployment steps:

1. Create a new Apps Script project and paste backend code in `Code.gs`.
2. Create `Index.html` and paste the frontend.
3. Enable Drive API via Services.
4. Deploy as Web App (Execute as: Me; Who has access: Anyone in organization).
5. Authorize permissions and share Web App URL with end users.

### 12.2 Required OAuth Scopes

Scopes include:

- `spreadsheets` - Read/write target Google Sheet
- `drive, drive.readonly` - Temp file handling and metadata
- `script.storage` - Save/retrieve Sheet URL via PropertiesService
- `gmail.send` - Optional upload notification emails

### 12.3 Deployment Configuration Summary

Table 10: Deployment Configuration

| Setting         | Value                  |
|-----------------|------------------------|
| Project Name    | SAP Data Automation    |
| Execute As      | Me (Developer Account) |
| Who Has Access  | Anyone in Organization |
| Deployment Type | Web App                |



| Setting                   | Value                           |
|---------------------------|---------------------------------|
| Version                   | v1.0 — Production               |
| Environment               | Google Apps Script Cloud        |
| External Hosting Required | None                            |
| Additional Infrastructure | None                            |
| Monthly Hosting Cost      | Rs 0 (Free on Google Workspace) |

## 13 CHALLENGES & SOLUTIONS

Table 11: Challenges and Solutions

| # | Challenge Faced                                    | Solution Implemented                                                                                   |
|---|----------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| 1 | Excel files cannot be read directly in Apps Script | Used Drive API to convert .xlsx/.xls to Google Sheets format, extracted data, then trashed temp files  |
| 2 | Financial year URL persistence across sessions     | Used <code>PropertiesService.getUserProperties()</code> storing URL and FY as key-value pairs          |
| 3 | FY auto-expiry needed every April 1st              | Compared saved FY against current FY and cleared stored values on mismatch                             |
| 4 | SAP date formats incompatible with JavaScript Date | Built <code>parseSAPDate()</code> handling multiple SAP date formats                                   |
| 5 | SAP European number formats breaking calculations  | Built <code>parseSAPCurrency()</code> to normalize commas/dots                                         |
| 6 | Large Excel files risking 6-minute timeout         | Documented 5MB guideline and advised splitting files                                                   |
| 7 | Base64 encoding large files caused memory spikes   | Used <code>FileReader.readAsDataURL()</code> and split to extract base64 string                        |
| 8 | Sheet tab auto-selection after dynamic population  | Used <code>Array.from(...).find()</code> to locate and select tab                                      |
| 9 | Submit button state management across async calls  | Tied <code>submitBtn.disabled</code> to success/failure handlers of <code>validateSheetAccess()</code> |

## 14 KNOWN LIMITATIONS

Table 12: Known Limitations

| # | Limitation                                        | Severity | Recommendation                                           |
|---|---------------------------------------------------|----------|----------------------------------------------------------|
| 1 | Apps Script max execution time is 6 minutes       | Medium   | Split files larger than 5MB before uploading             |
| 2 | Drive API must be manually enabled per deployment | Low      | Clearly document in setup steps                          |
| 3 | Only first selected category processed per file   | Medium   | Implement category loop in v2.0                          |
| 4 | No role-based access control within the app       | Medium   | Restrict deployment access to org domain (short-term)    |
| 5 | PropertiesService stores only one URL per user    | Low      | Acceptable for single-sheet-per-FY workflow              |
| 6 | No undo or rollback after data is written         | High     | Implement automatic backup tab before each write in v2.0 |
| 7 | No internet connection makes app unavailable      | Low      | Expected for cloud-hosted app                            |
| 8 | Apps Script free tier has daily execution limits  | Medium   | Monitor usage via Google Cloud Console                   |

## 15 FUTURE RECOMMENDATIONS

This section outlines future enhancements grouped by priority to guide the roadmap for version 2.0 and beyond.

### 15.1 High Priority Recommendations (4 items)

- Multi-category processing loop** Implement a true multi-category loop so a single uploaded file can be written once per selected category instead of only the first selected category, removing the need to re-upload the same file multiple times.
- Automatic backup and rollback mechanism** Before each write, automatically create a timestamped backup tab (e.g. Backup\_YYYYMMDD\_HHMM) and provide a simple “Restore previous upload” option.
- Role-based access and environment separation** Introduce RBAC (Admin, Power User, Standard User) for configuration and bulk operations; clearly separate DEV, UAT, and PROD with visual indicators and property namespaces.
- Large-file handling and resumable upload** Implement chunk-based/batch writing with configurable max rows per batch and graceful timeout detection plus resume-from-last-row behavior.

## 15.2 Medium Priority Recommendations (6 items)

5. Centralized configuration and master data management via a “Config” sheet or JSON in `PropertiesService`.
6. Enhanced audit log and reporting dashboard built on `AuditLog` and `UploadSummary`.
7. Notification and alerting framework with optional email alerts and user preferences.
8. Advanced validation rules for SAP data with reference lists and a `ValidationIssues` tab.
9. Improved UX for multi-file upload with structured summary panel and error-only CSV download.
10. Extended browser/device support documentation and in-app help/tooltips.

## 15.3 Low Priority Recommendations (6 items)

11. Theming and UI customization (logo, color theme, dark/light mode).
12. Template-based summary sheets for department-specific reporting.
13. Contextual in-app help and guided tours for new users.
14. Multi-language label support via externalized text.
15. Optional scheduled reminder emails for key periods (month-end, quarter-end).
16. Export of configuration and logs for audit in a one-click package.

# 16 PROJECT METRICS SUMMARY

## 16.1 Development Metrics

- 7 iterative phases from requirements to deployment.
- Stack: Google Apps Script backend, HTML5/CSS3/JS frontend, Drive API, SpreadsheetApp, `PropertiesService`.
- 15+ core backend functions including parsing, validation, and logging.
- 3 feature groups: Core, Smart/Automated, SAP-Specific.
- 10 bugs identified and fixed; 0 outstanding at go-live.

## 16.2 Testing Metrics (60 test cases, 100% pass rate)

- Total test cases: 60.
- Passed: 60; Failed: 0.
- Pass rate: 100%.
- Coverage: URL validation, access control, file handling, SAP parsing, metadata, FY logic, performance, mobile, error handling.

### **16.3 Feature Delivery Metrics (40 delivered, 16 pending v2.0)**

- Approx. 56 features/sub-features identified.
- 40 delivered in v1.0, covering all critical business & NFRs.
- 16 planned for v2.0, aligned with Section 12 roadmap.
- 100% of High-priority requirements for v1.0 implemented and tested.

### **16.4 Cost Analysis (Rs 0 total cost)**

- Infrastructure cost: Rs 0 (hosted on existing Google Workspace).
- Additional licensing cost: Rs 0 (no third-party infrastructure).
- Ongoing maintenance cost: minimal (scripts and config).

### **16.5 Business Impact Metrics (90% time saved)**

- Estimated  $\sim 90\%$  reduction in manual effort for monthly SAP-to-Sheets processing.
- Lower error rates via structured parsing, validation, and metadata tagging.
- Multi-department usability (Operations, Finance, HR, Logistics).
- Faster availability of data for MIS and decision-making.

## **17 SIGN-OFF AND APPROVALS**

The system has been developed, tested, and deployed as per the agreed scope and is ready for business use. The following checklist captures handover items to operations and IT.

### **17.1 Handover Checklist (11 items)**

1. Source code handover (Apps Script project with backend and HTML/JS/CSS).
2. Configuration and environment details (deployment config, environment separation).
3. Functional specification and requirements document.
4. Technical design and architecture notes.
5. User guide and SOPs for end-users.
6. Test cases and execution results (60/60 passed).
7. Known limitations and recommendations.
8. Access and permission matrix (users/groups and sheet access).
9. Support and escalation process definition.
10. Backup and data retention approach for key sheets.
11. Formal sign-off from business sponsor and IT, marking project as Completed.

## 18 CONCLUSION

### 18.1 Key Achievements (10 items)

1. End-to-end automation of SAP-to-Google Sheets data flow.
2. Robust URL validation and tab selection ensuring correct destinations.
3. User-friendly web interface with multi-file upload, drag-and-drop, and clear feedback.
4. Financial year awareness with automatic pre-selection and FY-based URL persistence.
5. Comprehensive SAP-specific handling (movement types, dates, currency, exports).
6. Metadata-enriched datasets with Section, Category, Year, Month, Timestamp columns.
7. Reliable audit and summary mechanisms via `AuditLog` and `UploadSummary`.
8. Strong performance metrics under concurrent load and across browsers.
9. Complete testing coverage with 60/60 tests passing and all bugs fixed.
10. Zero incremental infrastructure/licensing cost using existing Google Workspace.

### 18.2 Final Status Summary

Table 13: Final Status

| Area                     | Status / Metric                     |
|--------------------------|-------------------------------------|
| Project Status           | Completed                           |
| Core Features            | Implemented and tested              |
| Smart Features           | Implemented and tested              |
| SAP-Specific Features    | Implemented and tested              |
| Total Test Cases         | 60                                  |
| Test Pass Rate           | 100%                                |
| Outstanding Bugs         | 0                                   |
| Infrastructure Cost      | Rs 0                                |
| Time Savings             | ~ 90% reduction in manual effort    |
| Ready for Production Use | Yes – signed off by Business and IT |

### 18.3 Closing Statement

The SAP to Google Sheets Automation System has successfully achieved its objectives of eliminating manual data transfer, enforcing consistent metadata, and enabling faster, more reliable reporting across departments. With a stable v1.0 release in production, a clear roadmap for future enhancements, and formal sign-off from all stakeholders, this application is positioned to become the standard platform for SAP data ingestion into Google Sheets within the organization.